Self-avoiding effective strings in lattice gauge theories

Source-backed arXiv sample

1 Extracted Lists

The list environments below are extracted from arXiv LaTeX source and wrapped for pdfLaTeX validation.

i) it is composed, at large distances, by $D-2$ free bosons, describing the transverse displacements of the string;

ii) each bosonic field is compactified on a circle of radius R_t related to the finite thickness of the flux tube. This allows fermionizing the model, which in turn guarantees that the colour flux tube does not self-overlap;

iii) the boundary conditions on quark lines are dictated by the lattice gauge theory (LGT) and fulfil a sort of orthogonality constraint such that, whenever the quark and antiquark lines coincide, the effective string vanishes . This forces the adimensional radius $\nu = \sqrt{\pi\sigma}R_t$ to be exactly $\nu = 1/4$.